9/14/2026

BASICS OF PYTHON

Artificial Intelligence Lab

Instructor: Sir Mustajab

Submitted by: Wasif Ali

Reg id: 245187

Class: BSCS F24 B



Lab No 01

1. (1 point) Write a python program to print 'Welcome to my code!'.

Python Code:

```
[1]
✓ 0s
print("Welcome to my code!")
Welcome to my code!
```

Pseudocode:

START

OUTPUT "Welcome to my code!"

End

2. (1 point) Write a python program which declares one integer variable and other character. Assign your name to character variable and age to integer variable and print these variables in proper order.

Python Code:

```
[2]
✓ 0s
name = "Alex"
age = 20

print("Name:", name)
print("Age:", age)
Name: Alex
Age: 20
```

Pseudocode:

START

SET name = "Alex"
SET age = 20
OUTPUT name
OUTPUT age

END

3. (1 point) Write a python program which prompt user to enter student name, registration number, subject name, marks obtained and total marks and assign all these to proper datatypes. Calculate percentage of obtained marks and assign it to another variable. Print out student name, subject and marks obtained (marks obtained should be in percentage).

Python Code:

```
[3]
✓ 26s
student_name = input("Enter student name: ")
reg_num = input("Enter registration number: ")
subject_name = input("Enter subject name: ")
marks_obtained = float(input("Enter marks obtained: "))
total_marks = float(input("Enter total marks: "))

percentage = (marks_obtained / total_marks) * 100

print("\n--- Student Result ---")
print("Student Name:", student_name)
print("Subject:", subject_name)
print(f"Marks Obtained: {percentage:.2f}%")
```

```
Enter student name: Wasif
Enter registration number: 245187
Enter subject name: AI
Enter marks obtained: 100
Enter total marks: 100

--- Student Result ---
Student Name: Wasif
Subject: AI
Marks Obtained: 100.00%
```

Pseudocode:

START

INPUT student_name
INPUT reg_num
INPUT subject_name
INPUT marks_obtained
INPUT total_marks

SET percentage = (marks_obtained / total_marks) * 100

OUTPUT student_name
OUTPUT subject_name
OUTPUT percentage

END

4. (1 point) Write a `convertToFahrenheit()` function with a `degreesCelsius` parameter. This function returns the number of this temperature in degrees Fahrenheit. Then write a function named `convertToCelsius()` with a `degreesFahrenheit` parameter and returns a number of this temperature in degrees Celsius.

   Use these two formulas for converting between Celsius and Fahrenheit:

   - $\text{Fahrenheit} = \text{Celsius} \times \left(\frac{9}{5}\right) + 32$
   - $\text{Celsius} = (\text{Fahrenheit} - 32) \times \left(\frac{5}{9}\right)$

Pseudocode:

START

```
FUNCTION convertToFahrenheit(degreesCelsius)
  RETURN degreesCelsius * (9 / 5) + 32
END FUNCTION

FUNCTION convertToCelsius(degreesFahrenheit)
  RETURN (degreesFahrenheit - 32) * (5 / 9)
END FUNCTION

Main
  SET resultFahrenheit = convertToFahrenheit(25)
  SET resultCelsius = convertToCelsius(77)

  OUTPUT resultFahrenheit
  OUTPUT resultCelsius
```

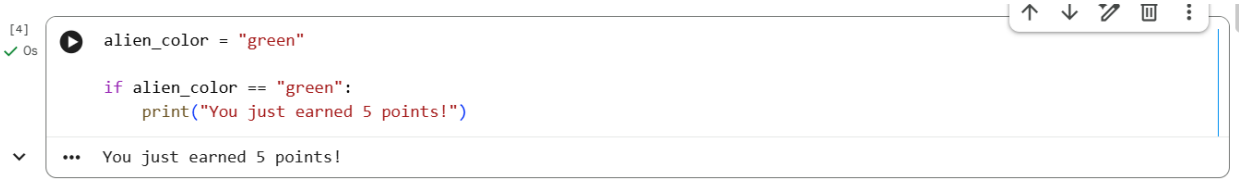
END

5. (1 point) Imagine an alien was just shot down in a game. Create a variable called `alien_color` and assign it a value of `'green'`, `'yellow'`, or `'red'`.

   - Write an `if` statement to test whether the alien's color is green. If it is, print a message that the player just earned 5 points.
   - Write one version of this program that passes the `if` test and another that fails. (The version that fails will have no output.)

Version 1 (Passes)

Python Code:



Pseudocode:

START

Step 1: Set alien_color = 'green'
Step 2: If alien_color equals 'green':
Display "You just earned 5 points!"
END

Version 2 (Fails)

Python Code:



Pseudocode:

START

Step 1: Set alien_color = 'yellow'
Step 2: If alien_color equals 'green':
Display "You just earned 5 points!"
END

6. (1 point) Modify the `if-else` chain from Task 5 into an `if-elif-else` chain:

   - If the alien is **green**, print a message that the player earned 5 points.
   - If the alien is **yellow**, print a message that the player earned 10 points.
   - If the alien is **red**, print a message that the player earned 15 points.

Python Code:

```
[7]
✓ 0s
alien_color = "green"

if alien_color == "green":
    print("You earned 5 points!")
elif alien_color == "yellow":
    print("You earned 10 points!")
else:
    print("You earned 15 points!")

You earned 5 points!
```

```
[8]
✓ 0s
alien_color = "red"

if alien_color == "green":
    print("You earned 5 points!")
elif alien_color == "yellow":
    print("You earned 10 points!")
else:
    print("You earned 15 points!")

You earned 15 points!
```

Pseudocode:

START

Step 1: Set alien_color = 'green'
Step 2: If alien_color equals 'green':
    Display "You earned 5 points!"
Else If alien_color equals 'yellow':
    Display "You earned 10 points!"
Else:
    Display "You earned 15 points!"

END

7. (1 point) Write an `if-elif-else` chain that determines a person's stage of life. Set a value for the variable `age`, and then:

   - If the person is **less than 2 years old**, print a message that the person is a **baby**.
   - If the person is **at least 2 years old but less than 4**, print a message that the person is a **toddler**.
   - If the person is **at least 4 years old but less than 13**, print a message that the person is a **kid**.
   - If the person is **at least 13 years old but less than 20**, print a message that the person is a **teenager**.
   - If the person is **at least 20 years old but less than 65**, print a message that the person is an **adult**.
   - If the person is **65 years or older**, print a message that the person is an **elder**.

Python Code:

[9]
0s

```
age = 25

if age < 2:
    print("The person is a baby.")
elif age < 4:
    print("The person is a toddler.")
elif age < 13:
    print("The person is a kid.")
elif age < 20:
    print("The person is a teenager.")
elif age < 65:
    print("The person is an adult.")
else:
    print("The person is an elder.")
```

```
The person is an adult.
```

Pseudocode:

START

Step 1: Set age = 25
Step 2: If age < 2:
    Display "The person is a baby."
  Else If age < 4:
    Display "The person is a toddler."
  Else If age < 13:
    Display "The person is a kid."
  Else If age < 20:
    Display "The person is a teenager."
  Else If age < 65:
    Display "The person is an adult."

Else:
Display "The person is an elder."
END

8. (1 point) Create a program that simulates how websites ensure that everyone has a unique username:

   - Make a list of five or more usernames called `current_users`.
   - Make another list of five usernames called `new_users`. Ensure that one or two of the new usernames are also in the `current_users` list.
   - Loop through the `new_users` list to check if each new username has already been used.
     - If a username has been used, print a message that the person will need to enter a new username.
     - If a username has not been used, print a message saying that the username is available.
   - Ensure that the comparison is **case insensitive**. If `'John'` has been used, `'JOHN'` should not be accepted. (To achieve this, create a copy of `current_users` containing the lowercase versions of all existing usernames.)

START

Step 1: Set current_users = ["admin", "Wasif", "Anisa", "Hamza", "Zainab"]
Step 2: Set new_users = ["WASIF", "Bilal", "ANISA", "Tariq", "Usman"]
Step 3: Convert all names in current_users to lowercase and save to current_users_lower
Step 4: For each user in new_users:
Convert user to lowercase
If lowercase user is in current_users_lower:
Display "Username is taken. Enter a new username."
Else:
Display "Username is available."
END

9. (1 point) Create a program that processes sandwich orders:

- Make a list called `sandwich_orders` and fill it with the names of various sandwiches.
- Create an empty list called `finished_sandwiches`.
- Loop through the list of sandwich orders using **while loop** and print a message for each order, such as: *I made your tuna sandwich.*
- As each sandwich is made, move it to the list of `finished_sandwiches`.
- After all the sandwiches have been made, print a message listing each sandwich that was made.

START

Step 1: Set sandwich_orders = ["tuna", "turkey", "club", "veggie"]
Step 2: Set finished_sandwiches = []

Step 3: While sandwich_orders is not empty:
    Remove first sandwich from sandwich_orders as current_sandwich
    Display "I made your " + current_sandwich + " sandwich."
    Add current_sandwich to finished_sandwiches

Step 4: Display "The following sandwiches have been made:"
Step 5: For each sandwich in finished_sandwiches:
    Display sandwich

END

10. (1 point) Write a `printASCIITable()` function that displays the ASCII number and its corresponding text character, from 32 to 126. (These are called the printable ASCII characters.)

    Your solution is correct if calling `printASCIITable()` displays output that looks like the following:

```
32
33 !
34 "
35 #

--more--
124 |
125 }
126 ~
```

    Note that the character for ASCII value 32 is the space character, which is why it looks like nothing is next to 32 in the output.

```
FUNCTION printASCIITable()
  For code from 32 to 126:
    Display code and CharacterFromASCII(code)
END FUNCTION

START

  Call printASCIITable()
END
```

11. (1 point) Write a program to print the output of a given number from last to first.

    - Take input from users in integer form.
    - Ensure that the input number is greater than zero.
    - Divide the number and print its result in integer form only.

```
START

  Step 1: Input num as integer
  Step 2: If num > 0:
        While num > 0:
```

```
            SET digit = num MOD 10
            Display digit
            SET num = num DIV 10
        Else:
          Display "Invalid input. The number must be greater than zero."
END
```